



# Data Flow Diagram

|               |                                            |
|---------------|--------------------------------------------|
| Date          | 10 July 2026                               |
| Team ID       | XXXXXX                                     |
| Project Name  | Voice-Based Concept Understanding Analyzer |
| Maximum Marks | 2 Marks                                    |

## Data Flow Diagram

The Data Flow Diagram (DFD) illustrates how information moves through the Voice-Based Concept Understanding Analyzer. It represents the interaction between external entities, system processes, data stores, and the flow of information from voice input to report generation. The diagram provides a high-level overview of how audio data is processed, analyzed, evaluated, stored, and finally presented to the user.

## DFD Symbol Legend

| Symbol                | Name            | Description                                                                                                                        |
|-----------------------|-----------------|------------------------------------------------------------------------------------------------------------------------------------|
| Oval / Rounded Shape  | External Entity | A person, organization, or external system that sends or receives information from the Voice-Based Concept Understanding Analyzer. |
| Rectangle with Number | Process         | Represents a processing activity such as Speech-to-Text, Semantic Analysis, Audio Feature Extraction, or Report Generation.        |

|                           |            |                                                                                              |
|---------------------------|------------|----------------------------------------------------------------------------------------------|
| Rectangle<br>(Data Store) | Data Store | Stores information such as uploaded audio files and generated reports for future retrieval.  |
| Labeled Arrow             | Data Flow  | Indicates the movement of information between external entities, processes, and data stores. |

## Data Flow Diagram Visualization

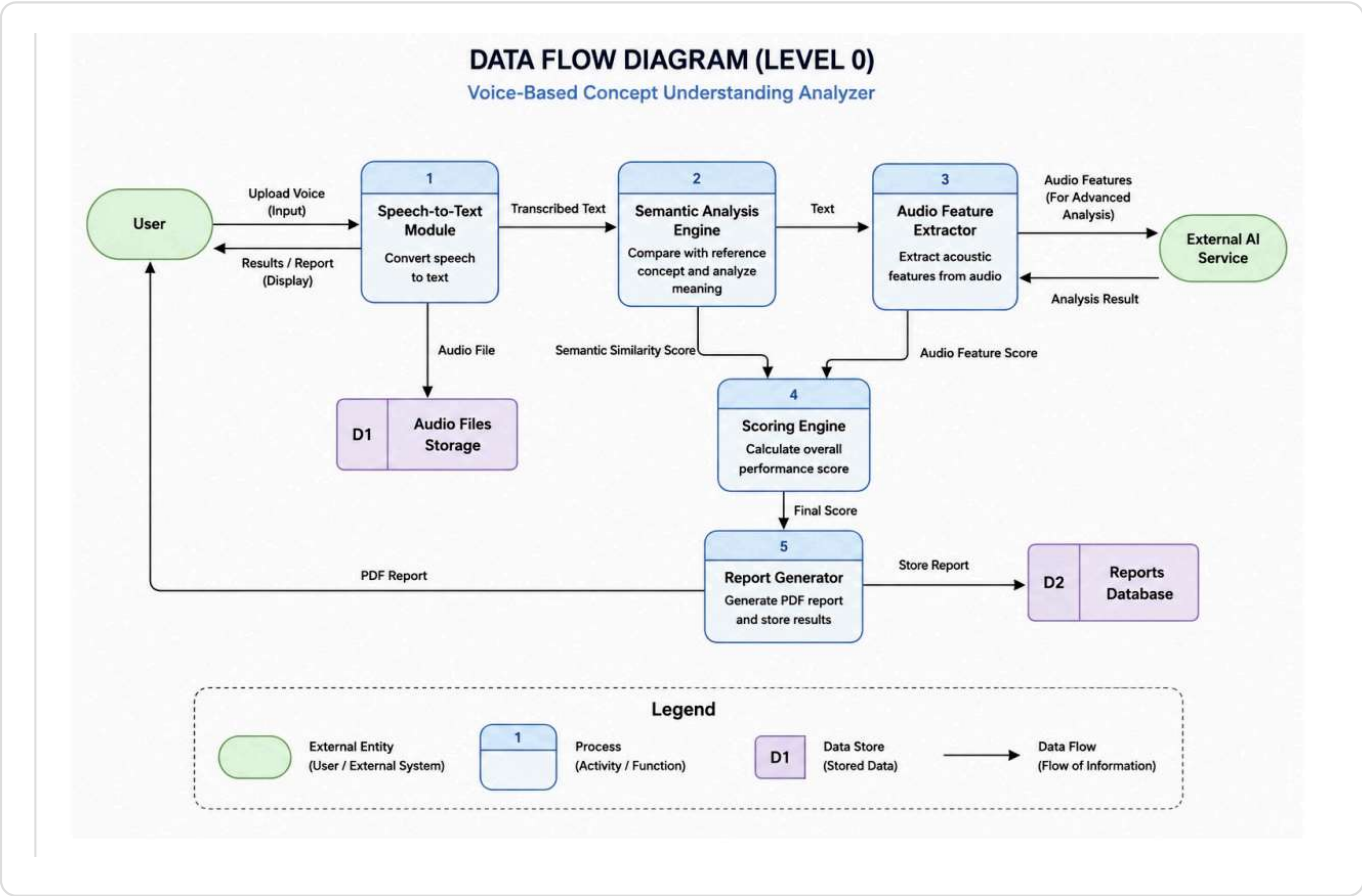


Figure 1: Data Flow Diagram of Voice-Based Concept Understanding Analyzer

## Description

The workflow begins when the user uploads a voice recording through the application. The Speech-to-Text Module converts the audio into text, which is forwarded to the Semantic Analysis Engine to compare the user's explanation with reference concepts. Simultaneously, the Audio Feature Extractor analyzes speech characteristics such as pitch, clarity, and fluency. The Scoring Engine combines semantic similarity and audio feature scores to calculate the final performance score. The Report Generator then

creates a PDF report, stores it in the Reports Database, and displays the final evaluation results to the user.